

Algebra II with Trigonometry

Chapter 5.2

Olympiad Problems

Saved tutoring response with MathJax rendering. Original PDF file:

[4401268_alg-2trig-h-chp-5-2-note-docx.pdf](#)

Problems

1. Without using a calculator, evaluate $\sin 75^\circ + \sin 15^\circ$ and explain any surprising symmetry in your answer.
2. Given that $\cos \theta = \frac{12}{13}$ for an acute angle θ , determine $\tan(2\theta)$ mentally, using identities and minimal computation.
3. Compute $(\cos 60^\circ + \cos 120^\circ + \cos 240^\circ)^2$ without a calculator.
4. The sum $S = \tan 15^\circ + \tan 45^\circ + \tan 75^\circ$ is an integer. Find its value, showing any clever simplification.
5. Let $x = \sin^2 30^\circ + \sin^2 45^\circ + \sin^2 60^\circ$. Find the exact value of x , using symmetry or patterns among the special angles.
6. For positive real numbers a and b , the triangle with vertices $(0, 0)$, $(a, 0)$, and (a, b) has angle θ at the origin. If $\sin \theta = \frac{b}{\sqrt{a^2+b^2}}$ and $\cos \theta = \frac{a}{\sqrt{a^2+b^2}}$, what is $\tan(2\theta)$ as a function of a and b ?